

Paena

On Chain Earned Wage Access (EWA) Platform

Context

EWA (Earned Wage Access) means workers can get part of the money they've already earned before their regular payday. Someone has worked 10 days, but payday is still 5 days away, with EWA, one can use a platform that analyzes your credit history and bank account through Plaid to determine your credit limit and lend that money to the user with a flat advancement fee.

We are building a direct to consumer EWA platform powered by stablecoins as a settlement layer. Our goal is to help anyone, especially in emerging economies, access their wages instantly and fairly without relying on predatory loans or delayed payroll cycles, achieving higher revenue through decentralized finance.

The Landscape

EWA's ICP is anyone living paycheck to paycheck around the world, whether banked or unbanked, working in formal or informal economies, when faced with short term liquidity needs. The concept is simple and sticky: give users early access to the money they will earn in the near future, instantly and with minimal friction.

The direct-to-consumer EWA model has already proven massive market fit. [Earnin](#), [Dave](#), and [Brigit](#) together serve tens of millions of users in the US, with Earnin alone processing over \$15 billion in wage advances and reportedly generating more than \$90 million in annual revenue.

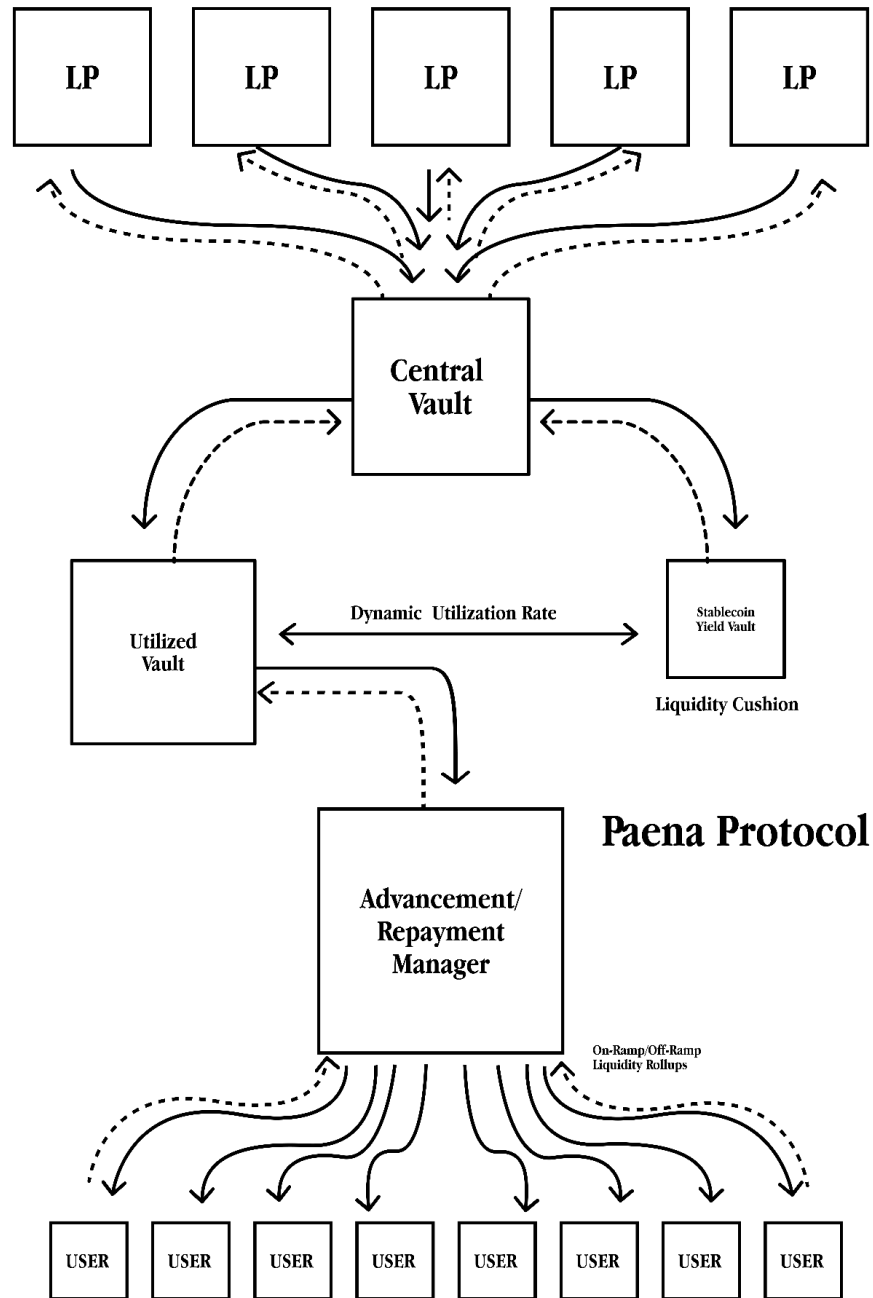
Yet these platforms rely on legacy financial rails (ACH, debit networks, and bank processors), which are slow, expensive, and margin-constrained. For every \$100 advanced, the cost of instant delivery via Visa Direct or Mastercard Send can reach 1.5–2%, and ACH failures add to losses and delays. To protect margins, EWA companies charge flat “tips” or fees that translate to effective **APRs of 100–400%**, especially on smaller advances.

The key lies in the high APR. Earnin, for example, was [sued](#) over average effective rates of around **284%**, highlighting both an ethical and structural flaw in the model. By removing intermediaries and reducing transaction costs, we can offer users significantly lower fees while still maintaining strong unit margins through on-chain decentralized finance.

Solution

Our solution brings earned wage access fully on-chain, replacing legacy ACH rails with stablecoin settlement for instant, low-cost transfers. Liquidity providers fund a vault that advances wages to users, earning yield from repayments and idle deposits. We sponsor on-ramp and off-ramp fees, allowing users to receive fiat payouts at near-zero cost while keeping the experience as simple as a traditional finance app.

Structure is as follows:



The liquidity structure operates through a single central vault that aggregates all deposits from liquidity providers. This vault acts as the primary source of funds for user wage advancements and simultaneously maintains a liquidity cushion to ensure stability during periods of heightened withdrawal or funding demand. The cushion functions as an internal safeguard, allowing the

system to satisfy user requests instantly without liquidating other assets or creating a bottleneck.

From the remaining capital that is actively utilized, funds flow to the advancement and repayment manager, which handles the disbursement of earned wage advances and the automated repayment process at the end of each user cycle. Once repayments are collected, the net proceeds, after accounting for minor default losses and the cost of on-ramping and off-ramping fiat, are split between the liquidity providers and the platform. Liquidity providers receive yield on both the deployed capital and the idle cushion, resulting in a blended return that combines active yield from deployed funds and passive yield from the liquidity reserve.

The platform's revenue derives from its share of the net yield generated by these deployments, with no intermediary banking partners or ACH drips involved, ensuring a fully on-chain flow with minimal friction and transparent capital circulation. The platform also drives revenue if LPs wish to withdraw their money instantaneously. We use that fee to ensure APR balance to all parties.

On the user side, they may advance at a reduced fee, a third to a fifth of other platforms. They receive it instantly or within minutes. At the end of the paycheck cycle, we use similar models to Earnin to directly debit to their bank account since we connect through Plaid to determine their credit limit.

We can also limit on-ramp and off-ramp fees by batching liquidity together and only operating fiat to crypto on the difference of inflows and outflows:

Assuming 0.05% on-ramp and off-ramp fee, without batching:

Fee per batch = $0.005 * (\text{Advances} + \text{Repayments})$ because we need to on-ramp/off-ramp every single transaction.

With batching:

Fee per batch = $0.005 * |\text{Advances} - \text{Repayments}|$ because we only need to on ramp and off ramp the difference between inflows and outflows since some repayments immediately fund advancements. This dramatically reduces on-ramp and off-ramp fees.

We charge a small advance fee averaging 50% APR. Liquidity providers earn approximately 20% APR from wage advances, while we retain the remaining 30% as platform revenue. In the United States, over 15 million workers living paycheck to paycheck rely on EWA. Assuming an average user advances \$3,000 per year, capturing just 2.2% of this market would generate \$100 million in annual platform revenue. Globally, \$31.9 billion was advanced, and capturing just 0.94% of that would generate \$100 million in revenue. Scaling to other countries would also be plausible since we are stablecoin-oriented.

The Math:

Running through this structure with a \$1M example pool. We try the best to adjust the APR rates with the tiered flat fee structure to maintain an APR around 50%, which is drastically lower than the 200-400% APR that other apps are charging. We also assume a 2% charge-off rate, as well as a 4% drag with on-ramping/off-ramping.

We used this [study](#) to determine that the average advancement period for EWA apps is 10 days, and we used a \$100 advancement as a sample advancement.

Table 3. Average Advance Amounts and Days to Repay by Lender

Company	All Lenders	Brigit	Cleo	Dave	EarnIn	FloatMe
Average Advance Amounts	\$79	\$58	\$39	\$133	\$83	\$20
Minimum Advance	\$10	\$25	\$20	\$20	\$10	\$10
Maximum Advance	\$500	\$280	\$250	\$500	\$300	\$50
Average Days to Repay	10.0	10.8	10.3	10.5	8.8	9.0

Source: CRL analysis of SaverLife data.

Gross yield on deployed capital:

After 2% charge-offs on a 50% APR product, net yield = 48%.

LPs receive 50% of that = 24% APR on deployed capital.

Blended yield across total pool:

Eighty percent of the pool is deployed, twenty percent sits in a 6% buffer.

$r_{LP, total} = 20.4\%$ APR

Net APR for LPs (after defaults):

$\approx 20.4\%$ APR on the total pool.

On a \$1,000,000 pool they make roughly \$204,000 per year.

Platform

Gross yield on deployed capital (before costs):

Same 24% APR on deployed capital (half of the 48% net yield).

On/off-ramp drag:

We sponsor fiat-crypto conversions at a 4% effective APR drag on deployed capital.

$r_{Platform, net} = 24\% - 4\% = 20\%$ APR on deployed capital

Pool-wide yield:

Since only 80% of the pool is deployed,

$r_{Platform, poolwide} = 20\% \times 0.80 = 16\%$ APR

Platform revenue on \$1M pool:

$\$1,000,000 \times 16\% = \$160,000$ per year

Users

We charge flat per-advance fees instead of a percentage of paycheck to stay compliant. We give users two options:

1. Express Fee (instantaneous)
2. Standard Fee (receive funds by the end of the day)

According to the [study](#), 70% of users choose Express even if they have to pay more, and we use that statistic to calculate our current APR earnings.

User Fee Structure (Targeted / Tiered)

Tier	Advance Amount	Express Fee	Standard Fee	Effective APR (10-day advance with \$100)
Tier 1	\$0–\$100	\$1.50	\$0.99	54.8% (fast) – 36% (standard)
Tier 2	\$100–\$200	\$3.00	\$1.99	54.8% (fast) – 36% (standard)
Tier 3	\$200–\$400	\$3.99	\$2.99	73% (fast) – 36% (standard)

So, a \$100 instant advance repaid in 10 days, the average repay period, corresponds to ~54.8% APR, far below the 300–900% APR equivalents seen in legacy EarnIn-style apps. CCheaper for the user, but also a healthy margin for us. For the same typical \$100, 10-day advancement order on Earnin, the user would have to pay \$3.99 at 145% APR.

In short: LPs earn about 20.4% APR net, the platform net 16% APR on the pool (20% on deployed) after covering on/off-ramp costs, and users effectively pay about 50% APR on average. Having this high of an APR is crucial in incentivizing LP to deposit with us. This yield level is higher than a majority of DeFi yield products on the market. It's also lower risk as all loans are subject to the user's pay period, and unlike Aave-type DeFi loans, which must be over-collateralized because they have no fixed payback period.

TAM

Our initial go-to-market focus will target low- to moderate-income workers who frequently face short-term cash flow challenges, as well as gig and hourly shift workers.

These include three major segments: gig workers such as rideshare and delivery drivers, hourly shift workers in retail, hospitality, and food service, and independent contractors such as freelancers and tradespeople. Together, these groups represent approximately 44.5 million workers in the U.S. labor force.

Not every worker will adopt EWA services, so realistic adoption rates are applied. Assuming 25 percent adoption among gig workers, 10 percent among hourly shift workers, and 8 percent among independent contractors, the addressable user base comes to about 4.9 million people. Based on existing EWA usage patterns, the average annual advance per user is estimated at \$4000 for gig workers, \$2500 for hourly workers, and \$3000 for contractors. Multiplying the number of users in each segment by their respective average advance amounts results in a total annual advance volume of roughly 14.3 billion dollars across all addressable users.

At a fee equivalent to a 50 percent annual percentage rate, which translates to roughly five percent of the total advance amount in short-term usage fees, the total annual revenue opportunity (TAM) equals about 715 million dollars. To achieve one hundred million dollars in annual recurring revenue, the platform would need to capture approximately 14% of this addressable market, which equates to around 666,000 active users. Each of these users would advance about three thousand dollars per year, generating an average of 150 dollars in annual fee revenue.

The bottom-up total addressable market for an on-chain EWA platform serving U.S. workers is approximately 715 million dollars in yearly fee revenue. Reaching one hundred million dollars in ARR would require engaging about two-thirds of a million users, or roughly thirteen to fourteen percent of the realistic addressable base. Gig workers are likely to be the most profitable initial focus, since they both advance larger sums and experience more frequent liquidity gaps, making them the strongest early adopters for instant pay solutions.

Globally, the opportunity is far larger. More than 1 billion people around the world live paycheck to paycheck, representing a massive population of workers who regularly face liquidity gaps and depend on immediate income access. If each of these workers advanced just \$1000 per year and the platform captured a modest 50% APR fee, that would represent a global earned wage access opportunity of 50 billion dollars in potential annual revenue. Even capturing only 1% of this global population, about 10 million workers, would translate into roughly 500 million dollars in yearly revenue. Onchain EWA can scale internationally, expanding from the U.S. market into emerging economies where wage delays, informal labor, and limited access to traditional banking systems make instant earned income access both impactful and inevitable.

The Team

Team includes a DeFi engineer running \$5 million dollars in TVL at an a16z startup and a ex-founder who scaled a consumer crypto product to top \$60M TGE on Virtuals.